# Technical Report: Doctor Management System

# Page 1 — System Overview

## 1. Introduction

The Doctor Management System is a web-based feature where the **Admin can add and manage doctors**, while **Patients can view available doctor information**. The purpose of this system is to make doctor information easy to manage and accessible to patients.

## 2. Main Users

The system has two main users:

- **Admin:** Adds and manages doctor information.
- **Patient:** Views doctor information and selects a doctor according to their needs.

## 3. System Workflow

**Admin → Add Doctor → Store Doctor Information → Patient Views Doctor Information**

The admin enters the doctor's details through an admin panel. The information is stored in the database. When a patient opens the doctor page, the system retrieves the available doctor information and displays it.

## 4. Doctor Information

The admin can enter information such as:

- Doctor ID
- Doctor Name
- Specialization
- Qualification
- Experience
- Phone Number
- Email
- Hospital/Clinic
- Available Days

- Available Time
- Doctor Image

# 5. Admin Module

The Admin module provides a form for adding doctors. After submitting the form, the system validates the information and saves it in the database.

Basic process:

1. Admin logs into the system.
2. Admin opens **Add Doctor**.
3. Admin enters doctor information.
4. System validates the data.
5. Data is inserted into the Doctor table.
6. A success message is displayed.
7. The doctor becomes available on the patient side.

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# Page 2 — Technical Implementation

# 6. Patient Module

Patients can open the **Doctors** page to see available doctors. The system fetches doctor records from the database and displays them in cards or a table.

Example information displayed:

**Dr. Ahmed Khan**

Specialization: Cardiologist

Qualification: MBBS, FCPS

Experience: 10 Years

Available: Monday–Friday

Time: 10:00 AM–2:00 PM

Patients can use this information to understand the doctor's specialization, experience, and availability.

# 7. Database Design

A `doctors` table can be created with the following fields:

| Field | Type | Description |
|---|---|---|
| doctor_id | Integer | Unique doctor ID |
| name | Text | Doctor name |
| specialization | Text | Medical specialization |
| qualification | Text | Doctor qualification |
| experience | Integer | Years of experience |
| phone | Text | Contact number |
| email | Text | Email address |
| clinic | Text | Hospital/clinic name |
| available_days | Text | Available days |
| available_time | Text | Available time |
| image_url | Text | Doctor image |

# 8. Technology Used

The system can be developed using:

- **Frontend:** Next.js / React
- **Styling:** CSS
- **Backend:** Next.js server-side functions
- **Database:** Supabase
- **Authentication:** Supabase Auth
- **Storage:** Supabase Storage for doctor images

# 9. Data Flow

```
Admin Login
     ↓
Add Doctor Form
     ↓
Validation
     ↓
Supabase Database
     ↓
Doctors Table
     ↓
Patient Doctor Page
     ↓
Doctor Information Displayed
```

# 10. Security and Validation

Only authorized administrators should be allowed to add or modify doctor information. Input validation should be applied to required fields such as doctor name, specialization, email, and contact number. Supabase Row Level Security (RLS) can be used to control database access.

# 11. Conclusion

The Doctor Management System provides a simple connection between the **Admin and Patient modules**. Admins can add accurate doctor information to the system, and patients can view that information through the website. Using **Next.js and Supabase** makes the system suitable for a modern, database-driven healthcare website.